Integration by Parts

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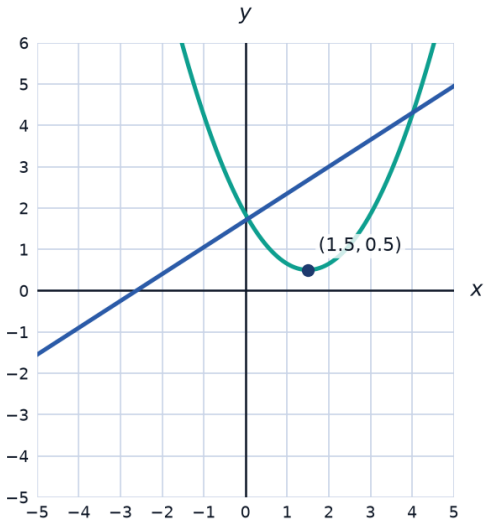
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The integration by parts formula**

Evaluate $\int xe^x\,dx$ using integration by parts.

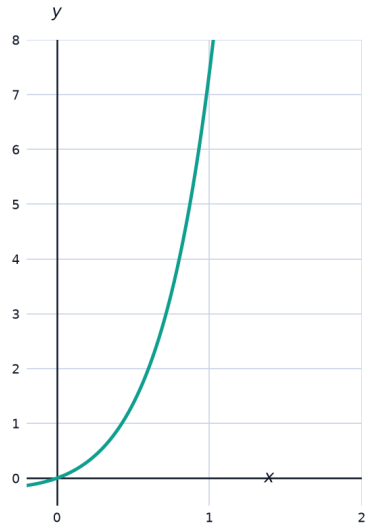
2 **[Foundational] • Target: Repeated and definite applications**

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.



3 **[Foundational] • Target: Repeated and definite applications**

Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.



4 **[Foundational] • Target: Choosing u and dv wisely (LIATE)**

Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

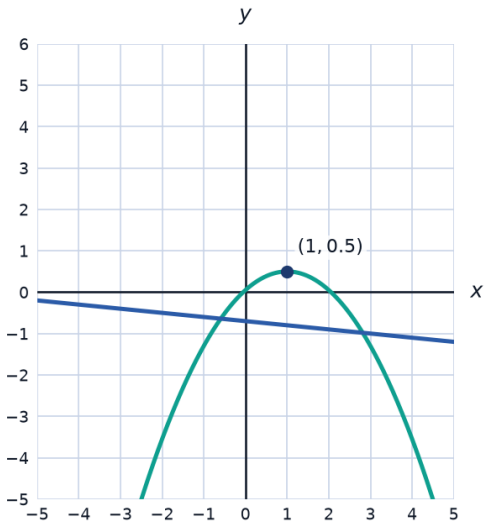
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5 [Foundational] • Target: The recurring-integral trick

Evaluate $\int e^x \sin x \, dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

6 [Foundational] • Target: Reduction formulas

Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x \, dx = x^n e^x - n\int x^{n-1} e^x \, dx$.



7 [Foundational] • Target: Reduction formulas

Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x \, dx$, confirming it matches an earlier result.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

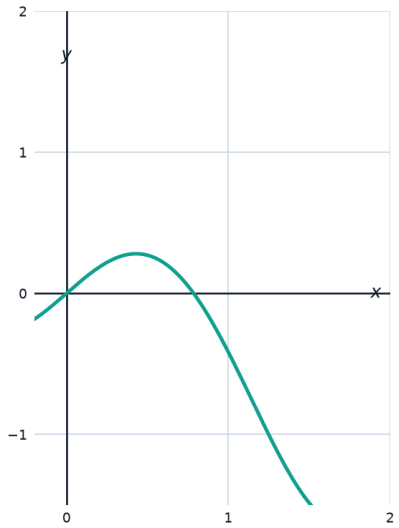
8 [Intermediate] • Target: Integration by parts with inverse trig functions

Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

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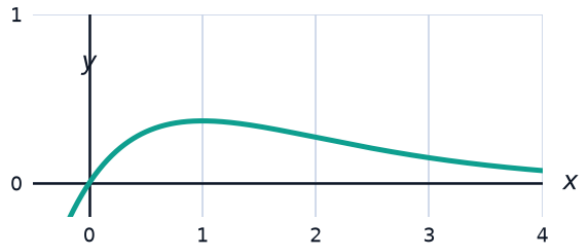
9 **[Intermediate]** • **Target: A capstone mixed-technique problem**

Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.



10 **[Intermediate]** • **Target: Visualizing an area found by integration by parts**

The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.



11 **[Intermediate]** • **Target: Choosing between substitution and parts**

For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

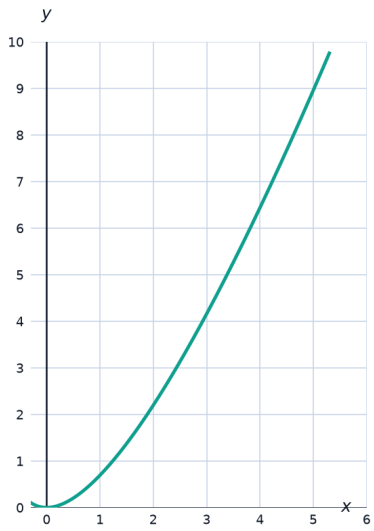
12 **[Intermediate]** • **Target: Choosing between substitution and parts**

Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

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13 [Intermediate] • Target: Choosing between substitution and parts

A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.



14 [Intermediate] • Target: The recurring-integral trick

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \ge 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \ge 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

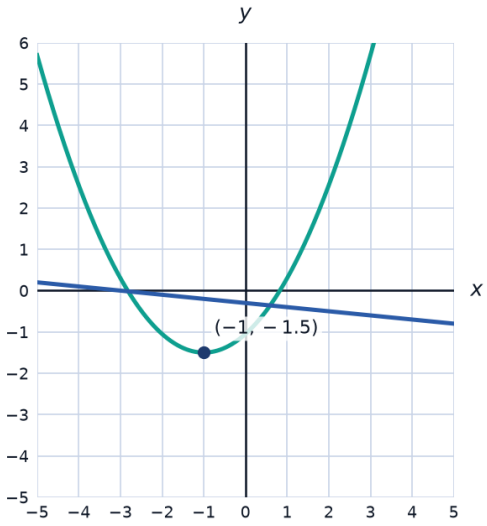
15 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 2.02 Continuity

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

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16 **[Challenging] • Target: Reduction formulas • Cross-Lesson: 3.01 The Derivative from First Principles**

[Unit 8 · Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging] • Target: Integration by parts with inverse trig functions • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^{\infty} xe^{-2x}\,dx$ or establish that it diverges.

18 **[Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Level 2 · MediumFind the equations of all horizontal and vertical tangent lines to the ellipse $4x^2 + 9y^2 = 36$.

19 **[Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Level 2 · MediumThe base of a solid is bounded by $y = \sqrt{x}$, the line $x = 9$, and the x-axis. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of the solid.

20 **[Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The integration by parts formula

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

22 [Foundational] • Target: Repeated and definite applications

Reduction formula: $I_n = \int \sin^n x\,dx$ satisfies:

a. $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$

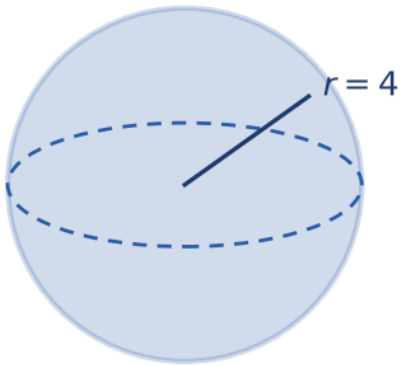
b. $I_n = \frac{\sin^{n+1} x}{n+1} + C$

c. $I_n = -n\cos x \sin^{n-1} x + C$

d. $I_n = I_{n-1} \cdot I_1$

23 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Volume between paraboloids $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$:



a. 4π

b. 8π

c. 16π

d. $32\pi/3$

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24 [Foundational] • Target: The recurring-integral trick

Consumer surplus formula (demand $p = D(q)$, equilibrium (q^*, p^*)):

a. $\int_0^{q^*} [D(q) - p^*]\,dq$

b. $\int_0^{p^*} D(q)\,dq$

c. $p^* q^* - \int_0^{q^*} D(q)\,dq$

d. $D(q^*) - p^*$

25 [Foundational] • Target: Definite integrals by parts

Reduction formula for $\int \sin^n x\,dx$ comes from IBP with $u =$

a. $\sin^n x$

b. $\cos x$

c. $\sin^{n-1} x$

d. x

26 [Foundational] • Target: Reduction formulas

Tabular integration (repeated parts): $\int x^3 e^x\,dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3

27 [Foundational] • Target: Integration by parts with inverse trig functions

Archimedes' principle as an integral: buoyancy force = weight of fluid displaced = ρg times:

a. surface area

b. volume of submerged body

c. pressure × depth

d. mass of body

28 [Foundational] • Target: A capstone mixed-technique problem

Surface area of revolution about x-axis: $S =$

a. $\pi \int_a^b f(x)\sqrt{1 + f'(x)}\,dx$

b. $2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx$

c. $2\pi \int_a^b [f'(x)]^2\,dx$

d. $\int_a^b f(x)dx$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Visualizing an area found by integration by parts**

Area of surface of revolution $r = f(\theta)$ about polar axis:

a. $2\pi\int f\sin\theta\sqrt{f^2 + f'^2}\,d\theta$

b. $\pi\int f^2\,d\theta$

c. $2\pi\int f\,d\theta$

d. $4\pi\int f\cos\theta\,d\theta$

30 **[Intermediate]** • **Target: Choosing between substitution and parts**

Gauss-Green formula in 2D: $\iint_\Omega abla\cdot\mathbf{F}\,dA =$

a. $\oint_{\partial\Omega}\mathbf{F}\cdot\mathbf{n}\,ds$

b. $\oint_{\partial\Omega}\mathbf{F}\cdot d\mathbf{s}$

c. $\iint_\Omega|\mathbf{F}|\,dA$

d. 0

31 **[Intermediate]** • **Target: The integration by parts formula**

$\int\sin^4 x\,dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

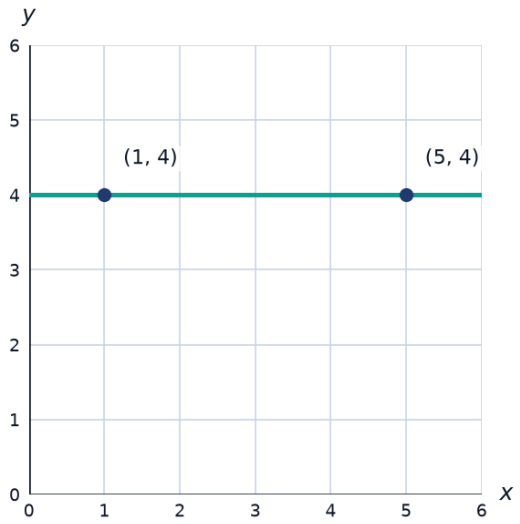
c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x\cos x}{4} + \frac{\sin x\cos x}{4} + \frac{x}{8} + C$

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32 [Intermediate] • Target: Repeated and definite applications

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.



a. 17

b. 21

c. 9

d. 11

33 [Intermediate] • Target: Choosing u and dv wisely (LIATE)

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

34 [Intermediate] • Target: The recurring-integral trick

Arc length in parametric form ($x = x(t), y = y(t)$):

a. $\int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$

b. $\int_a^b \sqrt{(dx/dt) + (dy/dt)}\, dt$

c. $\int_a^b |dy/dx|\, dt$

d. $\sqrt{x(b)^2 + y(b)^2}$

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35 [Intermediate] • Target: Definite integrals by parts

Kinetic energy of rotating rod (length L, mass M, angular velocity ω):

a. $\frac{1}{3}M\omega^2L^2$

b. $\frac{1}{6}M\omega^2L^2$

c. $\frac{1}{2}M\omega^2L^2$

d. $M\omega^2L^2$

36 [Intermediate] • Target: Reduction formulas

Force on a flat plate submerged at depth d, width $w(y)$, density ρ:

a. $\rho g\int w(y)(d-y)\,dy$

b. $\rho g dw$

c. $\rho w d^2$

d. $\int w\,dy$

37 [Intermediate] • Target: Integration by parts with inverse trig functions

Moments: $M_y = \rho\int_a^b x[f(x)-g(x)]dx$. Centroid $\bar{x} = M_y/M$ where $M =$

a. $\rho\int_a^b[f(x)-g(x)]dx$

b. $\rho(b-a)$

c. $\rho \cdot A$ (same as first)

d. M_y

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Probability of event $a \le X \le b$ for PDF f:

a. $\int_a^b f(x)\,dx$

b. $f(b)-f(a)$

c. $f(a)\cdot(b-a)$

d. $F(b)\cdot F(a)$

39 [Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Centroid of semicircular region radius R, $y \ge 0$:

a. $\bar{y} = R/2$

b. $\bar{y} = 4R/(3\pi)$

c. $\bar{y} = 2R/3$

d. $\bar{y} = R/\pi$

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40 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A cylindrical tank of radius 3 m and height 2 m is full of water. Write the work to pump all the water to a point 1 m above the top as $W = k\rho g$. Find k.

a. 36π

b. 54π

c. 36

d. 18π

41 [Challenging] • Target: The integration by parts formula • Cross-Lesson: 6.01 Related Rates

Work done lifting a 10 kg mass 4 m ($g = 10$ m/s²):

a. 400 J

b. 40 J

c. 4 J

d. 100 J

42 [Challenging] • Target: Repeated and definite applications • Cross-Lesson: 6.02 Optimization and Curve Sketching

Volume of revolution of $y = x^3$ about x-axis from 0 to 1:

a. π

b. $\pi/5$

c. $\pi/7$

d. $3\pi/7$

43 [Challenging] • Target: Choosing u and dv wisely (LIATE) • Cross-Lesson: 7.01 Integration by Substitution

Moment of inertia of disk radius R, mass M about its axis:

a. MR^2

b. $\frac{1}{4}MR^2$

c. $\frac{1}{2}MR^2$

d. $2MR^2$

44 [Challenging] • Target: The recurring-integral trick • Cross-Lesson: 8.01 Area Between Curves

Find the angle between $\mathbf{u} = \langle 1, 2, -4 \rangle$ and $\mathbf{v} = \langle 4, -2, 0 \rangle$.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{2}$

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45 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 8.02 Volumes of Revolution

Spring: force $F = kx$. Work stretching from 0 to d:

a. $kd^2/2$

b. kd

c. kd^2

d. $2kd$

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